

```
[9]: def ratrow():
      name='hari'
      for i in range(len(name)):
          for j in range(0,i+1):
              print(name[i],end=" ")
          print()
      ratrow()
```

```
h
a a
r r r
i i i i
```

```
[10]: def ratcoloumn():
        name='hari'
        for i in range(len(name)):
            for j in range(0,i+1):
                print(name[j],end=" ")
            print()
        ratcoloumn()
```

```
h
h a
h a r
h a r i
```

```
[11]: def ratupperrow():
        name='hari'
        for i in range(len(name)):
            for j in range(0,i+1):
                print(name[i].upper(),end=" ")
            print()
        ratupperrow()
```

```
H
A A
R R R
I I I I
```

```
[12]: def ratuppercoloumn():
        name='hari'
        for i in range(len(name)):
            for j in range(0,i+1):
                print(name[j].upper(),end=" ")
            print()
        ratuppercoloumn()
```

```
H
H A
H A R
H A R I
```

```
[13]: def ratlowerrow():
        name='HArI'
        for i in range(len(name)):
            for j in range(0,i+1):
                print(name[i].lower(),end=" ")
            print()
        ratlowerrow()

h
a a
r r r
i i i i
```

```
[14]: def ratlowercolumn():
        name='HArI'
        for i in range(len(name)):
            for j in range(0,i+1):
                print(name[j].lower(),end=" ")
            print()
        ratlowercolumn()

h
h a
h a r
h a r i
```

```
[17]: def latrow():
        name='hari'
        for i in range(len(name)):
            for j in range(len(name),i,-1):
                print(' ',end=' ')
            for k in range(i+1):
                print(name[i],end=' ')
            print()
        latrow()
```

```
h
a a
r r r
i i i i
```

```
[18]: def latcolumn():
        name='hari'
        for i in range(len(name)):
            for j in range(len(name),i,-1):
                print(' ',end=' ')
            for k in range(i+1):
                print(name[k],end=' ')
            print()
        latcolumn()
```

```
h
h a
h a r
h a r i
```

```
[19]: def latupperrow():
        name='hari'
        for i in range(len(name)):
            for j in range(len(name),i,-1):
                print(' ',end=' ')
            for k in range(i+1):
                print(name[i].upper(),end=' ')
            print()
        latupperrow()
```

```

        H
       A A
      R R R
     I I I I
```

```
[20]: def latuppercoloumn():
        name='hari'
        for i in range(len(name)):
            for j in range(len(name),i,-1):
                print(' ',end=' ')
            for k in range(i+1):
                print(name[k].upper(),end=' ')
            print()
        latuppercoloumn()
```

```

           H
        H A
       H A R
      H A R I
```

```
[21]: def latlowerrow():
        name='hArI'
        for i in range(len(name)):
            for j in range(len(name),i,-1):
                print(' ',end=' ')
            for k in range(i+1):
                print(name[i].lower(),end=' ')
            print()
        latlowerrow()
```

```

        h
       a a
      r r r
     i i i i
```

```
[22]: def latlowercoloumn():
        name='hArI'
        for i in range(len(name)):
            for j in range(len(name),i,-1):
                print(' ',end=' ')
            for k in range(i+1):
                print(name[k].lower(),end=' ')
            print()
        latlowercoloumn()
```

```

           h
        h a
       h a r
      h a r i
```

```
[23]: def pramidrow():
        name='hari'
        for i in range(len(name)):
            for j in range(len(name),i,-1):
                print('',end=' ')
            for k in range(i+1):
                print(name[i],end=' ')
            print()
        pramidrow()
```

```

        h
       a a
      r r r
     i i i i
```

```
[24]: def pramidcoloumn():
        name='hari'
        for i in range(len(name)):
            for j in range(len(name),i,-1):
                print('',end=' ')
            for k in range(i+1):
                print(name[i],end=' ')
            print()
        pramidcoloumn()
```

```

        h
       a a
      r r r
     i i i i
```

```
[25]: def factors():
        n=int(input("Enter the number:"))
        for i in range(1,n+1):
            if n%i==0:
                print(i)
        factors()
```

```

Enter the number: 80
1
2
4
5
8
10
16
20
40
80
```

```
[26]: def factors():
        n=int(input("Enter the number:"))
        for i in range(1,n+1):
            if n%i==0:
                print(i)
        factors()
```

```

Enter the number: 10
1
2
5
10
```

```
[8]: #BIGGEST NUMBER USING LAMDA FUNCTIONS
def bignum(a=80,b=100,c=90,d=60):
    num=lambda a,b,c,d:a if a>b and a>c and a>d else b if b>c and b>d else c if c>d else d
    print(num(a,b,c,d))
bignum(90,45,60,30)
```

90

```
[9]: #SMALL NUMBER USING LAMDA FUNCTIONS
def smallnum(a=80,b=100,c=90,d=60):
    num=lambda a,b,c,d:a if a<b and a<c and a<d else b if b<c and b<d else c if c<d else d
    print(num(a,b,c,d))
smallnum(90,45,60,30)
```

30

```
[12]: #SQUARE USING LAMBDA MAP FUNCTION
def square():
    num=[4,6,8,3,5]
    sq=list(map(lambda a:a**2,num))
    print(sq)
square()
```

[16, 36, 64, 9, 25]

```
[14]: #cube using lambda functions
def cube():
    num=[4,6,8,3,5]
    sq=list(map(lambda a:a**3,num))
    print(sq)
cube()
```

[64, 216, 512, 27, 125]

```
[15]: #cube root using lambda function
def cuberoot():
    num=[7,8,125,27,99]
    root=list(map(lambda a:a**(1/3),num))
    print(root)
cuberoot()
```

[2.3333333333333335, 2.6666666666666665, 41.666666666666664, 9.0, 33.0]